Section 5.5 U-substitution

$$\frac{d}{dx}\left[f(g(x))\right] = f'(g(x)) \cdot g'(x)$$

So, $$\int f'(g(x)) \cdot g'(x)\, dx = f(g(x)) + c$$

Ex ① $y = (x^2+1)^5$

$$y' = 5(x^2+1)^4 \cdot 2x$$

$$\int 5(\underbrace{x^2+1}_{\text{inner}})^4 \cdot \underbrace{2x}_{(\text{inner})'}\, dx$$

$$= (x^2+1)^5 + c$$

let u = inner function

du = ______ MUST show up!

Ex ② $\int (x^2+7)^{19} \cdot x\,dx$

let $u = x^2+7$

$dx \cdot \frac{du}{dx} = 2x \cdot dx$

$\frac{1}{2} \cdot du = \frac{1}{2} 2x \cdot dx$

$\frac{1}{2} du = x \cdot dx$

$\int u^{19} \cdot \frac{1}{2} du$

$= \frac{u^{20}}{20} \cdot \frac{1}{2} + C$

$= \frac{(x^2+7)^{20}}{40} + C$

ck

$\frac{d}{dx}\left[\frac{(x^2+7)^{20}}{40} + C \right]$

$\frac{20(x^2+7)^{19} \cdot 2x}{40}$

$= (x^2+7)^{19} \cdot x$ ✓

Ex ③ $\int \sqrt{3x+4}\ \underline{dx}$

let $u = 3x+4$

$dx \cdot \frac{du}{dx} = 3 \cdot dx$

only mult or divide by constants $\longrightarrow$ $\left[\frac{du}{3} = \frac{3 \cdot \underline{dx}}{3} \right]$

$\frac{du}{3} = dx$

$\frac{1}{3} \cdot \int \sqrt{u} \cdot du$

$\frac{1}{3} \int \frac{u^{1/2+1}}{\frac{3}{2}} du$

$\frac{1}{3} \cdot \frac{2}{3} \cdot u^{3/2} + c$

$\frac{2}{9}(3x+4)^{3/2} + c$

Ex ④ $\int x \cdot \sqrt{2x+4}\, dx$

$$u = 2x + 4 \xrightarrow{\text{solve for } x} \frac{u-4}{2} = x$$

$$du = 2 \cdot dx$$

$$\frac{1}{2} du = dx$$

$$\int \frac{(u-4)}{2} \cdot \sqrt{u} \cdot \frac{1}{2} du$$

$$\frac{1}{4} \int (u-4) \cdot u^{1/2} du$$

$$\frac{1}{4} \int \left(\frac{u^{3/2 \, +1}}{\frac{5}{2}} - \frac{4u^{1/2 \, +1}}{\frac{3}{2}} \right) du$$

$$\frac{1}{4} \left[\frac{2}{5} \cdot u^{5/2} - 4 \cdot \frac{2}{3} u^{3/2} \right] + C$$

$$\frac{1}{4} \left[\frac{2}{5} (2x+4)^{5/2} - \frac{8}{3} (2x+4)^{3/2} \right] + C$$

Ex 5 $\int \cos(8x)\,dx$

$$u = 8x$$

$$du = 8 \cdot dx$$

$$\frac{1}{8}du = dx$$

$$\frac{1}{8}\int \cos(u) \cdot du$$

$$\frac{1}{8}\sin(u) + C$$

$$\frac{1}{8}\sin(8x) + C$$

$\underline{\text{Ex}}$ ⑥ $\int \tan x \, dx$

$= \int \frac{\sin x}{\cos x} dx$

let $u = \cos x$

$du = -\sin x \cdot dx$

$-du = \sin x \cdot dx$

$= -\int \frac{du}{u} = -\int \frac{1}{u} \cdot du$

$= -\ln u + c$

$= -1 \cdot \ln(\cos x)^{-1} + c$

$= \ln(\sec x) + c$

$\underline{\text{Ex}}$ 7 $\displaystyle\int \frac{\ln x}{x}\,dx$

$\cancel{u = x}$

$\cancel{u = \frac{1}{x}}$

$\cancel{du = -\frac{1}{x^2}\,dx}$

$u = \ln x$

$du = \frac{1}{x}\cdot dx$

$$\int u^1 \cdot du$$

$$= \frac{u^2}{2} + c$$

$$= \frac{(\ln x)^2}{2} + c$$